Kigali Balance: Making Every Recovered, Reclaimed and Destroyed Kilogram of HFC Add Up

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1. The problem

HFC refrigerants are recorded in separate systems: customs quotas, technician logbooks, waste slips and carbon-credit registries. We found no system that checks whether the kilograms claimed as *recovered*, *reclaimed* or *destroyed* add up across borders and registries. This gap lets virgin or smuggled gas be sold as "reclaimed". It could also let the same destroyed kilogram be credited twice.

2. Why it matters

- **Super-pollutant.** One kilogram of R-404A, a common supermarket refrigerant, warms the planet as much as about 3.9 t of CO_2. Equipment "banks" hold an estimated 13–24 $GtCO_2e$ and leak about 1.5 $GtCO_2e$ per year [1]. Lifecycle management could avoid another 39 $GtCO_2e$ by 2050 [2].
- **End of life is where the bank escapes.** Japan has one of the most comprehensive lifecycle laws. Even so, in 2024 it recovered only 43% of the HFC left in scrapped equipment; the other 6,878 t (12.6 $MtCO_2e$) was released. Operators reported 575 t of leakage, while the national inventory estimated 6,283 t [3].
- **Illegal trade is growing.** EU HFC seizures rose from about 250 t in 2023 to about 670 t in 2024. Documented fraud includes labelling virgin gas as reclaimed, exceeding quotas and using false invoices. Illegal gas has reached Italian supermarkets and state-railway supply chains [4]. Prices are up about tenfold, and EU supply will fall by another 50% in 2027 [4].

3. Why "reclaimed" and "destroyed" are the weak points

Both labels now carry money.

- **"Reclaimed" escapes scarcity.** Since 2025 the EU has banned servicing refrigeration with *virgin* gas of GWP ≥ 2,500, such as R-404A. Labelled reclaimed gas stays legal until 2030 [5], and HFCs reclaimed inside the EU fall outside the shrinking quota [6]. Relabelling virgin gas as "reclaimed" therefore turns a banned, quota-limited product into a legal one.
- **"Destroyed" now earns credits.** Verra VM0016 v2.0 (Dec 2025) [7], Isometric (May 2026) [8], ACR, and a Taiwanese draft methodology (Jul 2026) [9] all credit HFC destruction. This market has a history: under the CDM, 19 HFC-23 projects out of about 2,000 received over half of all credits issued [10]. Isometric requires chain of custody for every container [8], but each registry can only check its own records.

4. What exists, and the gap

System	Covers	Missing
EU F-gas Portal + customs single window	Importer registration; quota at the border	Kilograms after first sale
Italy Banca Dati; France Trackdéchets	Technician interventions; recovered-waste slips	Not linked to each other or to supply. Italy still had the EU's largest seizures [4]
Türkiye FARAVET / EKOMVET	Quotas and imports; equipment and service records	Rules for waste F-gas are still a draft [11]
Japan RaMS	Electronic manifests from recovery to destruction	One country only; recovery stuck near 43% [3]
Recoolit; Verra / Isometric / ACR	Project tracking; credit issuance	No container-level uniqueness check across registries

Every segment has an owner, but no one owns the seams between segments. That is where the most profitable fraud happens.

5. Who needs this

Compliant reclaimers and distributors lose sales to cheap gas sold as "reclaimed". **Large buyers**, such as supermarkets and rail operators, need proof that their gas is legitimate. **Registries, verifiers and credit buyers** need proof that each destroyed kilogram came from real equipment and was credited only once. **Regulators** need warning signals that cut across segments. Technicians get no new paperwork: the system reuses records they already file.

6. Solution

Kigali Balance is a reconciliation layer. It ingests existing national and registry records rather than replacing them, and enforces one rule:

> **Every kilogram claimed as "reclaimed" or "destroyed" must trace back to a kilogram recovered from a real, registered piece of equipment.**

1. **Physical anchors.** Signed cylinder IDs; signed weighings at recovery, reclamation and destruction; signed lab composition certificates; equipment IDs from national registries.
2. **Conservation rules on a shared ledger.** (a) Every transfer balances: out equals in. (b) Kilograms recovered from a unit cannot exceed its registered charge. (c) Reclaimed output cannot exceed documented recovered input minus losses. (d) A party can only destroy what it holds. (e) Each destruction event is recorded once and can be claimed by only one registry before credits are issued, without exposing commercial data. (f) Gas charged into a unit in the last 180 days is quarantined: it can never be reclaimed or credited, so virgin gas cannot be laundered by charging it into a unit and recovering it the next day.
3. **AI that finds the implausible, not just the illegal.** (a) Mass-balance anomalies, such as reclamation yields or sales that outrun documented inflows. (b) Leakage gaps between expected and reported top-ups per unit, prompted by Japan's tenfold gap. (c) Recovery quantities or compositions that do not match a unit's type, age and service history. The AI flags cases for human review and never sanctions anyone automatically.

7. Why a shared ledger, and its limits

Inside one country with a strong regulator, a central database is enough; Japan's RaMS proves it. A shared ledger is justified only where **no single trusted operator spans all the parties**: cross-border trade between Türkiye and the EU, several independent carbon registries, and competing reclaimers. In those settings, conservation is enforced when each record is written instead of in an annual audit. Lying then requires collusion among independent counterparties.

The ledger has two honest limits:

- **It cannot see inside a cylinder.** Physical truth comes from weighing and lab sampling.
- **It cannot catch gas that never enters it.** Instead it makes legitimate gas *provable*, so buyers can demand provenance. It also stops illegal gas from being laundered into the legal market under a "reclaimed" label, which is that gas's most profitable exit.

8. Impact and alignment

- **How it cuts emissions.** Recovery rises only when someone pays for recovered kilograms, through credits, extended producer responsibility (EPR) schemes or a premium for reclaimed gas. Payment attracts fraud unless those kilograms are verifiable. *Illustration:* each percentage point of Japan's disposal recovery rate is worth about 120 t of HFC, or about 0.22 $MtCO_2e$ per year [3].
- **Adoption without a new mandate.** Participants join to qualify for credits or a reclaimed-gas premium. A US rule requiring QR-code cylinder tracking was struck down in court and then withdrawn after industry pushback.
- **COP31 and Türkiye.** Super-pollutants, including HFCs, were named "a central pillar of implementation on the road to COP31" at a June 2026 event attended by the COP31 President [12]. Türkiye rebuilt its F-gas regime in October 2024 [13]. Kigali Balance links FARAVET, EKOMVET and end-of-life records, giving compliant Turkish suppliers provable provenance in trade with the EU.
- **Taiwan.** Taiwan's HFC regulation (Feb 2025) and its draft destruction-credit methodology (Jul 2026) [9] need exactly this evidence.

9. What we do not claim

We are **not the first to track cylinders**: commercial tools, Japan's RaMS and a 2023 New Zealand blockchain project already do. Our contribution is cross-organisation conservation and cross-registry uniqueness. Double counting between destruction credits and national phase-down accounting is **undocumented**, so we treat it as a design safeguard, not a proven problem. The demo uses **simulated data only**.

References · [1] EIA, *ODS and HFC Banks*, 2026 · [2] UNEP guide citing TEAP 2024, Jul 2026 · [3] Japan MOE, OEWG48 side-event deck, Jul 2026 · [4] EIA, *Fakes, Fraud and F-Gases*, Apr 2026 · [5] Reg. (EU) 2024/573 Art. 13 · [6] German Environment Agency (UBA), F-gas Regulation overview · [7] Verra VM0016 v2.0 · [8] Isometric HFC & ODS Destruction Protocol v1.0 · [9] Taiwan MOENV HFC Management Regulation, 2025-02-25; PTS News, 2026-07-22 · [10] Carbon Market Watch, HFC-23 background paper · [11] iklim.gov.tr, draft waste F-gas circular · [12] CCAC, Jun 2026 · [13] Resmî Gazete No. 32693, 2024-10-15